COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	24/09/2026	**DATE SUBMITTED:**	24/09/2026
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MIDTERM **SKILLS EXAM**

1. PROCEDURE OUTPUT

Sample output (students are allowed to add more steps as needed.)
Task 1: Prepare the Server

```
tasks:
  - name: Task 1 - Prepare the Servers (Ubuntu)
    ansible.builtin.apt:
      update_cache: yes
      upgrade: dist
    when: ansible_os_family == "Debian"

  - name: Task 1 -  (CentOS)
    ansible.builtin.dnf:
      name: "*"
      state: latest
      update_cache: yes
    when: ansible_os_family == "RedHat"
```

```
- name: Task 1 - both
  ansible.builtin.package:
    name:
      - wget
      - tar
      - curl
      - net-tools
    state: present
```

```
TASK [Task 1 - Prepare the Servers (Ubuntu)] *
skipping: [192.168.56.12]
ok: [192.168.56.9]

TASK [Task 1 - Prepare the Servers (CentOS)] *
skipping: [192.168.56.9]
ok: [192.168.56.12]

TASK [Task 1 - both] ************************
ok: [192.168.56.12]
changed: [192.168.56.9]
```

Analysis: For task 1, what I did was I updated as well as upgraded both the ubuntu and centos by separating them through when condition. Packages were also installed for both.

Task 2: Create the Prometheus User

```
#
- name: Task 2 - Create prometheus system user
  ansible.builtin.user:
    name: prometheus
    system: yes
    shell: /bin/false
    create_home: no
```

```
TASK [Task 2 - Create the Prometheus User] *
changed: [192.168.56.9]
changed: [192.168.56.12]
```

Analysis: Here, I have created the user for prometheus. The /bin/false is used for denying the user to access the shell.

Task 3: Create Prometheus Directories

```
#
- name: Task 3 - Create Prometheus Directories
  ansible.builtin.file:
    path: "{{ item }}"
    state: directory
    owner: prometheus
    group: prometheus
    mode: '0755'
  loop:
    - /etc/prometheus
    - /var/lib/prometheus
```

```
TASK [Task 3 - Create Prometheus Directories] ********
changed: [192.168.56.9] => (item=/etc/prometheus)
changed: [192.168.56.12] => (item=/etc/prometheus)
changed: [192.168.56.9] => (item=/var/lib/prometheus)
changed: [192.168.56.12] => (item=/var/lib/prometheus)
```

Analysis: Here, the /etc/prometheus and /var/lib/prometheus were created as directories. Mode 0755 was used here for the groups and others to not have an access to write.

Task 4: Download Prometheus

```
  #
  - name: Task 4 - Download Prometheus
    ansible.builtin.unarchive:
      src: "https://github.com/prometheus/prometheus/releases/download/v3.14.0/prometheus-3.14>
      dest: /tmp
      remote_src: yes
```

```
TASK [Task 4 - Download Prometheus]
changed: [192.168.56.12]
changed: [192.168.56.9]
```

Analysis: Here, the prometheus source download is downloaded to the /tmp.

Task 5: Install the Prometheus Binaries

```
- name: Task 5 - Install the Prometheus Binaries
  ansible.builtin.copy:
    src: "/tmp/prometheus-3.14.0.linux-amd64/{{ item }}"
    dest: "/usr/local/bin/{{ item }}"
    owner: prometheus
    group: prometheus
    mode: '0755'
    remote_src: yes
  loop:
    - prometheus
    - promtool
```

```
TASK [Task 5 - Install the Prometheus Binaries] *
changed: [192.168.56.9] => (item=prometheus)
changed: [192.168.56.12] => (item=prometheus)
changed: [192.168.56.12] => (item=promtool)
changed: [192.168.56.9] => (item=promtool)
```

Analysis: Here, the binaries of prometheus are installed. Mode 0755 was given once again.

Task 6: Install the Prometheus Configuration Files
Task 7: Validate the Configuration

```
#
- name: task 6-7
  ansible.builtin.copy:
    src: "/tmp/prometheus-3.14.0.linux-amd64/prometheus.yml"
    dest: /etc/prometheus/prometheus.yml
    owner: prometheus
    group: prometheus
    mode: '0644'
    remote_src: yes
```

```
TASK [task 6-7] *********
changed: [192.168.56.9]
changed: [192.168.56.12]
```

Analysis:

Task 8: Create the systemd Service (OUT OF TIME)

```
#
- name: Task 8
  ansible.builtin.copy:
    dest: /etc/systemd/system/prometheus.service
    content: |
      [Unit]
      Description=Prometheus Time Series Collection and Processing Server
      Wants=network-online.target
      After=network-online.target

      [Service]
      User=prometheus
      Group=prometheus
      Type=simple
      ExecStart=/usr/local/bin/prometheus \
        --config.file=/etc/prometheus/prometheus.yml \
        --storage.tsdb.path=/var/lib/prometheus
      Restart=always

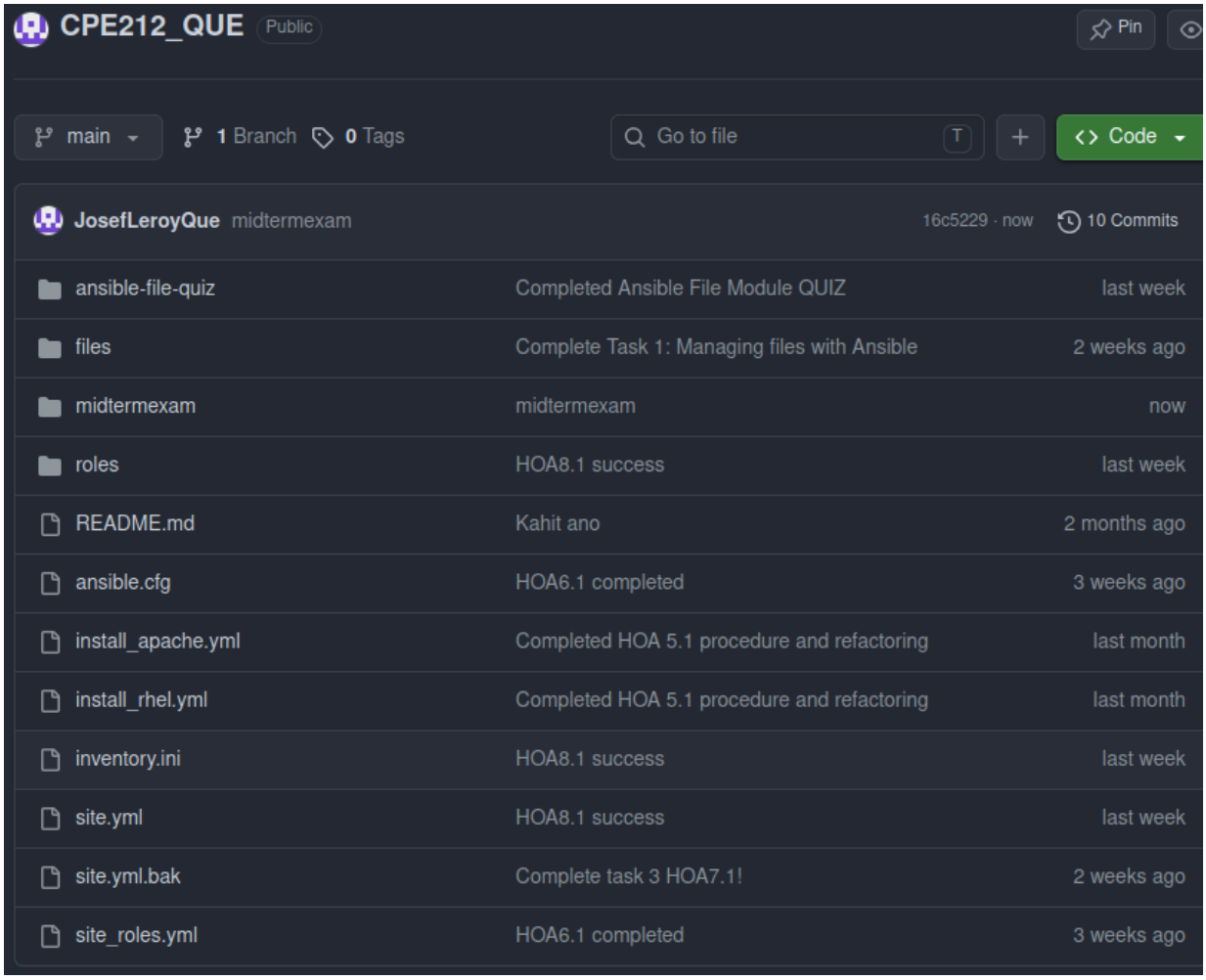
      [Install]
      WantedBy=multi-user.target
    mode: '0644'
```

Task 9: Start and Enable Prometheus (OUT OF TIME)

```
#
- name: Task 9
  ansible.builtin.systemd:
    name: prometheus
    state: started
    enabled: yes
    daemon_reload: yes
```

Task 10: Verify Prometheus
Task 11: Access Prometheus Through a Browser

Git commit:



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Purpose of Use: **N/A**

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